SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	---------
Project Name	Rising Waters
Maximum Marks	3 Marks

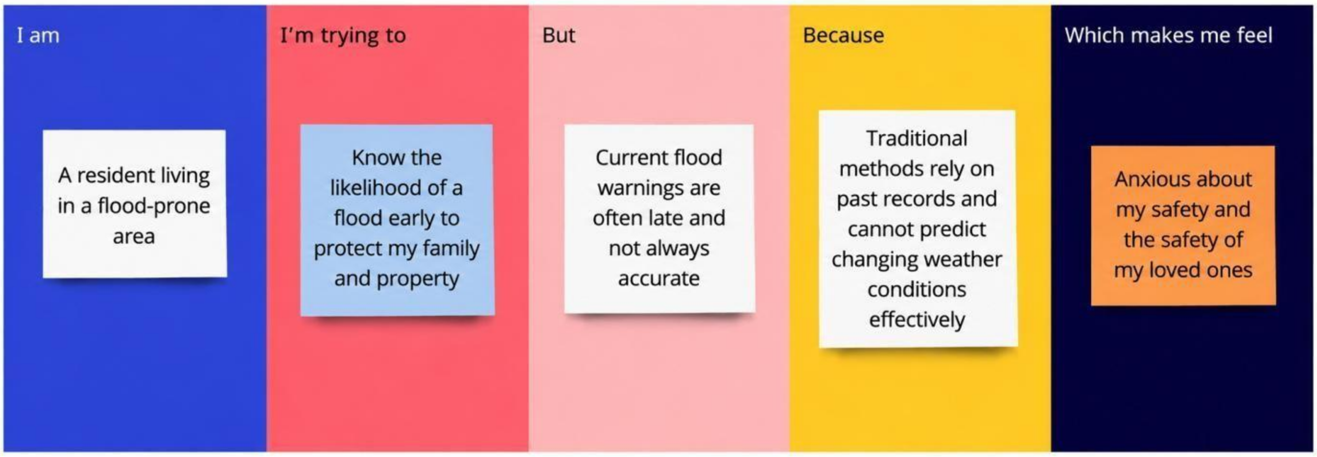
Define Problem Statements:

Floods are one of the most frequent and devastating natural disasters, causing loss of human lives, damage to infrastructure, agricultural losses, and economic disruption worldwide. With increasing climate change and unpredictable weather patterns, accurately forecasting floods has become more challenging. Traditional flood prediction methods often rely on manual analysis and historical records, which may not provide timely or accurate predictions.

There is a need for an intelligent, data-driven system that can analyze weather parameters such as annual rainfall, cloud cover, and seasonal rainfall to predict the likelihood of flooding. Such a system would enable government agencies, disaster management authorities, and local communities to receive early warnings, improve emergency preparedness, and make informed decisions.

The Rising Waters project addresses this challenge by developing a machine learning-based flood prediction system. The system trains multiple classification models, including Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost, to analyze historical weather data and identify flood risk. The best-performing model is integrated into a Flask web application, allowing users to enter weather parameters and instantly receive flood predictions. This solution aims to enhance disaster preparedness, reduce flood-related losses, and support effective resource planning through accurate and timely flood risk assessment.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A disaster management officer responsible for flood-prone regions.	Predict flood risks early and issue timely warnings to protect people and property.	Married with no co-applicant income.	Self-employed with a good credit history.	Optimistic about loan approval.
PS-2	A meteorologist monitoring weather conditions.	Analyse rainfall and weather data to forecast floods accurately.	Large amounts of weather data are difficult to analyze manually.	Traditional methods are time-consuming and less efficient.	Frustrated and under pressure to provide accurate forecasts.
PS-3	A resident living in a flood-prone area.	Receive early flood alerts to protect my family and belongings.	I often receive warnings too late.	Existing alert systems may not provide timely or location specific information.	Anxious about my family's safety and property.

PS-4	An environmental researcher studying flood patterns.	Develop an accurate flood prediction model using machine learning.	Building a reliable prediction model requires high-quality data and testing.	Different algorithms produce different results and require evaluation.	Motivated to improve prediction accuracy and disaster preparedness.